How Should We Read ViT Attention Maps? A Fast-Weight Perspective

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Abstract

What can a bright background patch in a vision transformer's attention map tell us? Holding the model and image fixed, we changed only how we drew the map. In both DeiT checkpoints, weighting received attention by each patch's own projected value length improved alignment with foreground masks and human annotations; deletion gains stayed below our practical threshold. On pet images, assigning the same lengths to other patches removed the enrichment gain in DeiT-S. Moving scores to decoded positions lowered all three metrics in all three models. Restricting the readout to the final classification query better ranked classifier interventions in both DeiTs, but foreground AUROC fell in DeiT-S and rose in DeiT-B.

1 Introduction

Deep-layer attention can highlight background patches. Adding register tokens can absorb some unusually high-norm background activations [11]. What changes if we keep the image and model fixed and change only how we read the map?

For a fixed image, each attention head acts as a two-layer fast-weight network with one hidden unit per source token [31]. For one head, the attention coefficient tells us how strongly a query reads a source. Multiplying that coefficient by the length of the source's projected value gives the length of the vector it returns to that query (Fig. 1). Norm-based analyses already distinguish these quantities [19, 36]. Whether that vector length predicts foreground, human annotations or classifier sensitivity is an empirical question.

Drawing on hidden-unit analysis [4], we ask three questions about a bright patch: Is it distinctive to the image? Does its position reflect where the token's content came from? How large a vector does it return? Matched controls test the readout each question suggests. We then ask whether the gain from value weighting survives moving the lengths to other patches, and whether a map for the classifier should pool the same queries as a foreground map.

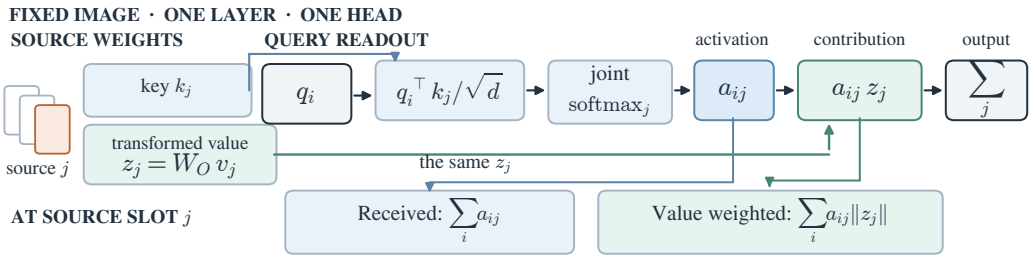


Figure 1: One head for a fixed image. Activation a_{ij} scales source j's output weight z_j. The two readouts sum either activation or the magnitude of the resulting vector (Eq. 1).

Preprint.

2 Related work

Activation and the returned vector. In a ViT [13], attention sets the coefficient of a source's value vector. Fast-weight memory [27, 28], Hopfield retrieval [26] and MiTA's two-layer representation [31] describe input-dependent weights and their readout. Norm-based analyses already distinguish attention coefficients from output magnitudes in language models [19] and visual attribution [36]. We ask what happens to foreground rankings when the same lengths are assigned to different patches.

Position and information origin. Token identity [7] and grid information [25] can survive contextual mixing. Rollout and relevance propagation trace paths across layers [1, 9, 10]; GlobEnc, ALTI and TokenTM account for additional transformations [15, 22, 34]. We test a simpler placement rule: assign a score to decoded grid positions. Registers and no-op heads help explain some high-norm outliers [6, 11, 18], whereas spatial priors can produce recurring saliency [20]. These alternatives motivate different controls.

What should a map explain? Attention can agree with annotations without explaining a prediction [17, 32]. Filtering aims to improve map agreement [3]; CAM targets a class score [35], and Value Zeroing measures contextual mixing [23]. Parameter dependence, supervision and occlusion choices also affect evaluation [2, 5, 30]. We evaluate the same rankings against masks and an intervention on the classifier path.

3 Methods for reading a frozen attention head

For query i, source j and head h, let $z_j^h = W_O^h v_j^h$ be the transformed value. With key dimension d, the attention output before residual addition is

$$o_i = b_O + \sum_{h,j} a_{ij}^h z_j^h, \qquad a_{ij}^h = \mathrm{softmax}_j(q_i^{h\top} k_j^h / \sqrt{d}). \tag{1}$$

For a fixed image, $k_j^h/\sqrt{d}$ and z_j^h act as the input and output weights of a two-layer network in q_i^h, with the softmax coupling its activations across all sources. Received attention assigns $s_j = \sum_{h,i} a_{ij}^h$ to patch j. Summing over all queries, including CLS, gives one score per patch on the 196-patch grid.

Specificity. A bright patch may reflect a recurring spatial pattern. We estimate a mean map on separate calibration images, measure the spatial variance it explains on test images, and subtract it from each test map. Subtracting a centred Gaussian scaled to the mean map's spatial standard deviation controls for a radial prior.

Location. After several layers, a token contains information from other patches. We train a linear probe to decode its grid index and redistribute s_j over the probe's predicted positions. A blur with the same mean squared displacement tests whether this outperforms smoothing alone. The probe decodes a grid index. Its accuracy tests whether position is recoverable; it does not trace content back to input pixels.

Output magnitude. A strongly attended source can return a short vector. We first measure attention on high-norm tokens: their norm entering the block, before LayerNorm, exceeds three times the image median. One control simply zeros their scores. *Value weighting* instead uses the returned vector at every patch, $c_j = \sum_{h,i} a_{ij}^h \|z_j^h\|_2$, without a threshold. For one query–source pair,

$$\Big\| \sum_h a_{ij}^h z_j^h \Big\|_2 \le \sum_h a_{ij}^h \|z_j^h\|_2.$$

The left side retains cancellation between heads; value weighting sums the right side over queries. These terms describe the attention branch before residuals, MLPs and subsequent token mixing.

4 Experiments on three reading assumptions

We compare layers 9–12 of frozen DeiT-S, DeiT-B [29] and DINO ViT-S [8]. Enrichment measures foreground overlap above a same-radius baseline on ImageNet-S (about 12,150 images) [16]; ClickMe compares agreement with own-image and other-image human maps (17,408 images) [14, 21]. Deletion measures how patch removal affects top-1 retention (2,000 images). We specified two comparisons in advance: each change against its matched control, and the combination against received attention (App. A). We require an absolute change of at least 0.02 (alignment) or 0.05 (deletion) and a paired 95% interval excluding zero.

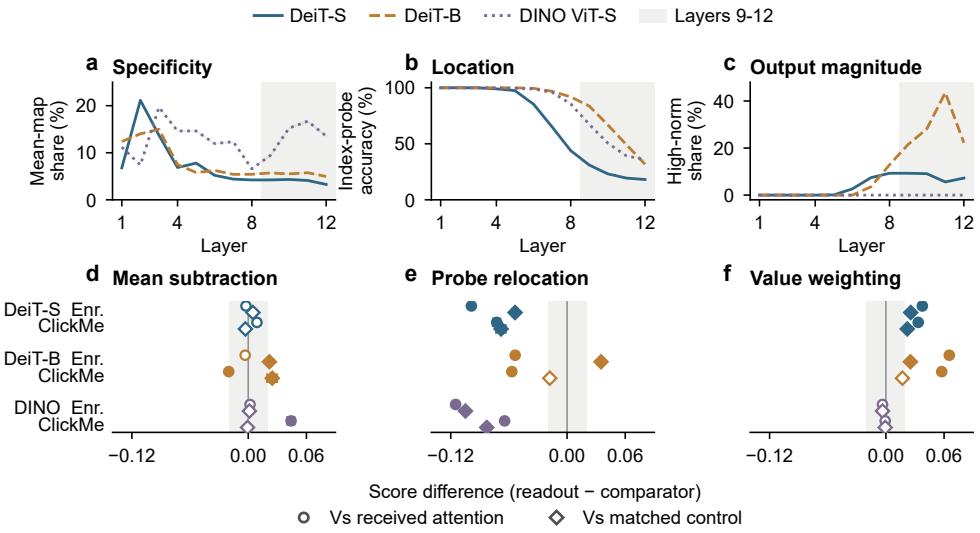


Figure 2: Does the phenomenon occur (top), and does changing the readout help (bottom)? Circles compare with received attention; diamonds compare with Gaussian subtraction, matched blur and zeroing high-norm scores, respectively. Bars show paired 95% intervals; filled points meet the decision rule. Grey shading marks layers 9–12 (top) or ±0.02 (bottom). Deletion: Table B.3.

Is a recurring bright spot informative? Across layers 6–12, the mean map explained about 3–6% of spatial variance in the DeiTs and up to 17% in DINO (Fig. 2a). In DINO, subtracting the mean raised ClickMe by 0.044, but subtracting a centred Gaussian gave a similar gain. In DeiT-B, mean subtraction beat Gaussian subtraction on all three metrics yet lowered ClickMe by 0.020 relative to received attention. A recurring pattern was not, by itself, a reason to remove it.

Where should a token's score go? At layer 12, grid-index accuracy was 18%, 32% and 36% in DeiT-S, DeiT-B and DINO (chance: 0.5%). Yet redistributing scores by decoded positions lowered all nine deep-layer scores. It also did worse than matched blur in DeiT-S and DINO, although it beat blur on DeiT-B enrichment (+0.035). Decodability did not justify relocation.

What does an active source send? At layers 9–12, high-norm patches received up to 9% of attention in DeiT-S and 44% in DeiT-B (Fig. 2c). Value weighting improved both alignment scores in both DeiTs. In DeiT-S, enrichment rose by 0.038 and ClickMe rose by 0.033. In DeiT-B the gains were 0.065 in enrichment and 0.058 on ClickMe. Weighting beat zeroing high-norm patches in all but DeiT-B ClickMe. Deletion rose by 0.017 in DeiT-S and 0.029 in DeiT-B: both intervals excluded zero, but neither gain reached 0.05. In DINO, where high-norm patches received no attention in these layers, value weighting changed each score by at most 0.004. For scale, DeiT-S enrichment was 0.178 with value weighting and 0.204 with GAE, which uses class gradients and all 12 layers (Table B.1).

Do the changes combine? Combining weighting, relocation and mean subtraction improved only DeiT-B ClickMe, against our prediction of four alignment gains. DeiT-B enrichment was comparable; both DeiT-S scores fell. Value weighting alone helped: did it matter which source received which length?

5 Two exploratory tests of what a readout explains

Which source gets which length? The placement test asks whether the same lengths would help if assigned to different patches. On 383 Oxford-IIIT Pet test crops with both foreground and background [24], two controls preserve each DeiT-S head's attention and set of lengths at each layer. One borrows the spatial rank order of lengths from another image; the other shuffles lengths only among patches at the same radius. Same-ring enrichment measures foreground alignment at matched radii; AUROC compares foreground with background regardless of radius.

Table 1: DeiT-S, layers 9–12, on 383 pet crops. ΔEnrichment is the row minus value weighting (own lengths) (paired 95% intervals).

Readout	Enrichment	ΔEnrichment [95% CI]	AUROC
Received attention	0.141	−0.019 [−0.023, −0.015]	0.788
Value weighting (own lengths)	0.160	—	0.817
Same lengths in donor rank order	0.137	−0.022 [−0.026, −0.019]	0.787
Same lengths shuffled within radius	0.138	−0.021 [−0.025, −0.018]	0.793
Own lengths, mean attention	0.120	−0.040 [−0.048, −0.032]	0.757

Reassigning lengths erased the enrichment gain over received attention (Table 1). Conversely, keeping lengths in place but averaging each head's received attention across patches lowered enrichment by 0.040. Both spatial patterns mattered. The gain could reflect two cues to foreground at the same patches; these comparisons do not isolate a benefit specific to multiplication.

Which receiving query matters? The placement tests pooled all queries. If the target is the classifier rather than the foreground, should every query still count? Only the final classification token (CLS) reaches the classifier, so we compared all queries with CLS alone against both targets. On 128 of the pet images, we held attention fixed in block 12, removed each patch's value term from the CLS update in turn, and recomputed the remaining classifier path. For each patch, we measured the absolute change in the logit margin between the original top two classes.

Using only CLS improved intervention ranking in both checkpoints (Table 2), as expected for an intervention on the CLS pathway. The masks tell a different story. In DeiT-S, foreground AUROC fell by 0.091. In DeiT-B, tested on the same images with both outcomes specified in advance, AUROC rose by 0.024 [0.013, 0.036]; the change in same-ring enrichment was uncertain ($+0.005$ $[-0.005, 0.013]$). Matching the query to the classifier did not consistently improve foreground ranking.

Table 2: Layer-12 value-weighted rankings, same 128 images. Intervention ρ: mean Spearman with CLS margin effects. Differences change only the query set. Full scores and intervals: App. E.4.

Model	Readout	Intervention ρ	Mask AUROC	Enrichment
DeiT-S	Value weighting, all queries	0.705	0.769	0.127
	Value weighting, CLS only	0.747	0.678	0.095
	CLS − all queries	$\mathbf{+0.041}$	$\mathbf{-0.091}$	$\mathbf{-0.033}$
DeiT-B	Value weighting, all queries	0.652	0.685	0.084
	Value weighting, CLS only	0.734	0.710	0.088
	CLS − all queries	$\mathbf{+0.082}$	$\mathbf{+0.024}$	$\mathbf{+0.005}$

6 Discussion and conclusion

A bright patch marks a source that received attention. Moving scores to decoded positions lowered all nine model–metric scores. Only value weighting improved both alignment metrics in both DeiTs, and reassigning the lengths erased the enrichment gain on pets. To interpret a map, specify what each patch score measures and which queries it pools, then test the ranking against the outcome the map is meant to explain.

Limitations and ethical considerations. Placement was tested in DeiT-S; receiver comparisons cover two checkpoints, one layer and 128 images. An intervention on a contextualised token cannot identify causal input pixels. Object size affects enrichment, and ClickMe may overlap training data.

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A Experimental details

Readout names in the archive. R0 denotes received attention; C1, mean subtraction; C2, probe relocation; and C3, weighting by projected value length. The combined readout (R_all) applies C3, then C2, then C1. M1–M3 identify diagnostic measurements; P1–P7 identify entries in the original registration.

Models. All models are frozen ViTs with 16×16 patches on 224×224 inputs. The three baseline backbones come from timm [33]. *DeiT-S and DeiT-B* [29]: `deit_small_patch16_224.fb_in1k` and `deit_base_patch16_224.fb_in1k` (resize to 248, centre crop 224: a crop ratio of 0.9). *DINO ViT-S/16* [8]: `vit_small_patch16_224.dino` with the official DINO linear head (resize to 256, centre crop 224: a crop ratio of 0.875). *DeiT-S with a test-time register* [18]: the activations of eight feed-forward units in layers 4 and 5 are moved to one added register token, which is left out of every map, like CLS. The units were selected on 500 ImageNet-S images, where they reduce the number of high-norm patch tokens (counted per layer) by 92.75% and keep 99.0% of the predictions; on the other 11,919 images the reduction is 91.5%, and ImageNet accuracy stays at 79.8%. For DeiT-B the same search found no such units (44.9% reduction with 32 units), so we report the register for DeiT-S only. *Top-k training*: the DeiT-S checkpoint released with MiTA [31] (25 summary slots, $k = 25$), with its own preprocessing (crop ratio 0.875). It differs from our DeiT-S in training and preprocessing, so its row in Table B.1 is not a controlled comparison; its deletion score is undefined because its summary slots pool fixed regions of the full grid. Both model-side changes are tested at size S only; in a preliminary comparison with an earlier protocol, top-k training scored 0.011 lower in object enrichment than its baseline at size S but 0.037 higher at size Tiny, so the result should not be read beyond DeiT-S.

Attention conventions. With x the output of the block's first layer norm, projections include the checkpoint's biases:

$$q_i^h = W_Q^h x_i + b_Q^h, \quad k_j^h = W_K^h x_j + b_K^h, \quad v_j^h = W_V^h x_j + b_V^h.$$

Equation 1 includes the output bias b_O and describes the attention branch before residual addition, with dropout inactive at evaluation. W_O^h is the head's slice of the output projection. C3 assigns the per-head vector magnitudes to patches; the shared output bias is not assigned to any patch. The fixed-image two-layer representation uses the coupled softmax activation, rather than an elementwise nonlinearity.

Data. *Execution test*: 5,000 randomly drawn ImageNet validation images [12]. *ImageNet-S* [16]: all 12,419 validation images with segmentation masks; stored enrichment is finite for 12,150 (12,144 for DINO). Historical records include all-foreground grids scored zero. Appendix E gives the cohort sensitivity check. Deletion uses a fixed random subset of 2,000. *ClickMe* [14, 21]: the 17,408-image validation split released with Harmonization (17 `val-*.tfrecords` shards). This split name does not identify the original ImageNet split of each image. Images are resized from 256×256 to 224×224 without cropping; heatmaps are resized to 64×64, blurred with a 10×10 Gaussian ($\sigma = 10$), then resized to 224×224 (bilinear interpolation). *Held-out set*: 4,000 ImageNet validation images excluded from map-quality and execution evaluation; 2,000 train probes and 2,000 provide mean maps, probe-test accuracies and blur calibration. Our near-duplicate check matched 2 of 4,002 candidate ImageNet validation images to ClickMe records; this does not establish every ClickMe image's training-set membership. Possible overlap with model training images limits generalisation claims.

Diagnostics. All three are computed per layer. *Specificity*: with $\bar{s}$ the mean received-attention map of the held-out images and Var_j the variance over the 196 patches of one image, the share explained is $1 - \sum_x \mathrm{Var}_j(s - \bar{s}) \,/\, \sum_x \mathrm{Var}_j(s)$, summed over the ImageNet-S images x. *Location*: the top-1 accuracy of the linear position probe on the 196 patch tokens of each of 2,000 held-out images. *Contribution*: the share of received attention on high-norm tokens, on the ImageNet-S images. The high-norm threshold (three times the median, defined below) is an operational definition fixed before the experiments; we did not test other thresholds. The correction itself uses no threshold.

Corrections and matched controls. Scores are computed per layer and summed over layers 9–12 without per-layer rescaling, except for the C1 control's calibration described below. In the three

baseline models, all 197 queries contribute to the readout, which scores only the 196 patch slots. Negative C1 scores are retained for ranking, without clipping. *Mean-map subtraction* computes $\sum_\ell (s_\ell - \bar{s}_\ell)$ using the 2,000 calibration images. With $S = \sum_\ell s_\ell$ and $\bar{S} = \sum_\ell \bar{s}_\ell$, its control is $S - G\ \mathrm{sd}_j(\bar{S})/\mathrm{sd}_j(G)$, where G is a centred isotropic Gaussian ($\sigma = 3.5$ patches). Calibration thus uses the *summed* mean map. *Probe relocation* computes $r_p = \sum_j s_j P_\ell(p \mid x_j)$, preserving total score. One linear 196-way classifier per model/layer is trained on standardised tokens after the first layer norm: cross-entropy, Adam, learning rate 0.01, 30 epochs, token minibatches of 4,096, seed 0. Feature means/standard deviations use only the 2,000 training images; standard deviations are floored at 10^{-6}. Its Gaussian blur control is a row-normalised kernel $K_{jp} \propto \exp(-\|p-j\|^2/(2\sigma_\ell^2))$. The width (0.7–2.9 patches) matches

$$D_\ell = \frac{1}{2000N} \sum_{x,j,p} P_\ell(p \mid x_j)\|p - j\|^2$$

on calibration images, averaging uniformly over source patch indices. ClickMe uses the same crop-trained probe and blur on resized full images, introducing a potential geometry mismatch. *Value weighting* computes $c_j = \sum_{h,i} a_{ij}^h \|W_O^h v_j^h\|_2$, a sum of per-head norm-weighted terms related to Kobayashi et al. [19]. Taking each head's magnitude before summing omits cancellation between heads; residual addition, subsequent normalisation and later-layer computation are outside this score. The control zeros received attention at high-norm patches: the residual-stream block-input norm *before* the first layer norm must exceed three times the image's median patch norm, using the midpoint median. C3 itself has no high-norm threshold. *All three corrections*: value weighting, probe relocation, then subtraction of the held-out mean of the resulting map, per layer before summation.

Metrics. *Same-ring object enrichment.* A patch is valid if it contains at least one non-ignored pixel, and is an object patch if at least half of its non-ignored pixels belong to the image's class mask. We select as many top-scoring patches as there are object patches. For every ring of patches at the same distance from the grid centre, we take the object rate among its selected patches minus that among all its patches, and average over rings, weighted by their selected patches. Invalid patches are excluded from selection and ring baselines. Ties are broken with the fixed seeded perturbation used in the original runs. A purely radial map scores zero in expectation over tie-breaking. *Image-specific ClickMe alignment.* The map is upsampled to 224×224 and correlated (Spearman, over pixels) with the image's own human map; we subtract its correlation with the previous image's human map within each batch (`roll(1)`; batch size 64). For an image-independent map, these differences average to zero over the paired set because the control permutes the same human maps; individual-image differences need not be zero. This removes fixed-prior alignment on average, without guaranteeing invariance to a prior mixed into an image-dependent map. *Deletion score.* Patches are ranked once by the original map. At removal budgets $0, 1, 2, 4, 8, 16, 32, 64, 98, 147$, we record the binary indicator that the original top-1 prediction is preserved. Tokens are dropped after positional embedding; CLS and any register remain. We use an SRG-style LIF–MIF area difference [5], adapting the value function from class probability to top-1 preservation. Areas use trapezoidal integration over removed-patch fraction $b/196$, from 0 to $147/196$. *Validity of the metrics.* A random map, a centred Gaussian and the mean map all score within 0.004 of zero on enrichment and on ClickMe alignment (the two image-independent maps have exactly zero mean on the latter, up to numerical precision); on deletion the centred Gaussian scores 0.110, 0.099 and 0.140 (DeiT-S, DeiT-B, DINO ViT-S), which is why every correction is compared with a matched control. *Statistics.* One value per image; paired image-bootstrap with 1,000 resamples, seed 0 and 95% percentile intervals. A difference is *higher* (*h*) or *lower* (*l*) if its interval excludes zero and its absolute mean reaches 0.02 (0.05 for deletion), and *comparable* (*c*) otherwise. The comparable category also covers imprecise differences. Intervals are unadjusted for multiple comparisons. Resampling holds checkpoints, probes and the within-batch ClickMe pairing fixed, excluding uncertainty from training, probe fitting or alternative pairings. Because the control reuses human maps within each batch, ClickMe score rows are not independent pairings.

GAE reference. The gradient-based reference [10] targets the unmodified model's predicted top-1 logit and uses all 12 attention layers. Starting with $R = I$, each layer updates $R \leftarrow R + \mathrm{mean}_h[(\nabla_{A^h} f_c \odot A^h)_+]R$, then takes the final CLS-to-patch row. It is class-specific and spans all layers, whereas R0/C1–C3 are class-agnostic and sum layers 9–12. Its scores provide descriptive context rather than a matched single-component ablation.

Registered predictions. The protocol was frozen before formal runs, after smoke tests on 32–64 images and earlier observations disclosed in `docs/preregistration.md`. The original main prediction (P3) was that the three changes together would beat received attention in four supervised-model alignment comparisons. Its rule assigns "holds" to four higher results, "partly holds" to one–three, and "does not hold" to none. One higher, one comparable and two lower results therefore receive the archived label "partly holds"; they do not support the predicted general improvement. Primary comparisons were the combination versus received attention and each change versus its control, with no direction predicted for the latter (P7). Other individual comparisons with received attention were descriptive, except the P4 negative controls. Prior execution and diagnostic observations had been seen, so P1 and P2 were replications. P5 reports last-layer recovery descriptively. Diagnostic explanations and supplementary mechanism analyses were post hoc.

Table A.1: Registered predictions and outcomes (centre-controlled metrics: enrichment and ClickMe alignment).

	Prediction	Outcome
P4a	Value weighting is comparable to received attention in DINO ViT-S on all three metrics	holds (3 of 3 cells)
P4b	Relocation at layers 1–4 is comparable to received attention on the centre-controlled metrics in all models	holds (6 of 6 cells)
P3	All three changes together are higher than received attention on the centre-controlled metrics in DeiT-S and DeiT-B	partly holds (1 higher: DeiT-B ClickMe; 1 comparable; 2 lower)
P6	Neither model-side change is higher than DeiT-S received attention on the centre-controlled metrics	holds (4 of 4 cells)
P1	Execution agreement rises with k and exceeds that of random sets at every k	holds in all models
P2	Probe accuracy falls with depth; the share of attention on high-norm tokens peaks in layers 6–12 of the supervised models and is below 1% in every layer of DINO ViT-S	holds in all models

Why test a register and MiTA checkpoint? These are model-side alternatives to changing how a frozen backbone is read. A test-time register moves selected feed-forward activations to an added token, reducing high-norm patch artefacts. The released MiTA checkpoint changes attention and training. Neither is an ablation of the two-layer representation, which also describes our unchanged baseline. P6 predicted that neither alternative would exceed the alignment margin at size S; all four comparisons were not higher. MiTA's different training and preprocessing restrict interpretation of that comparison.

Compute and code. The original experiments used approximately 19 NVIDIA L4 GPU hours. The original paper snapshot provides the registered protocol, configurations and fixed splits in `docs/preregistration.md`, `configs/` and `data/splits/`. The revised source includes later protocols, scripts and individual scores in `experiments/`. The README gives separate commands for rebuilding the displayed results and running inference.

B Additional results

We first report absolute scores and execution checks, then the paired differences used in Fig. 2. Additional analyses test whether map changes track high-norm attention and whether the maps depend on trained parameters.

Table B.1: Complete scores for the original tests, layers 9–12 unless indicated. Arrows mark higher/lower scores versus received attention (DeiT-S for model-side rows). Diamonds indicate higher scores than the matched control. Bold indicates higher than both. Untagged tested cells are comparable. GAE uses gradients and all 12 layers. Dashes indicate unavailable comparisons.

	DeiT-S			DeiT-B			DINO ViT-S		
Reading	Enr.	Click.	Del.	Enr.	Click.	Del.	Enr.	Click.	Del.
Received attention	0.140	0.157	0.236	0.080	0.107	0.169	0.199	0.208	0.285
Value weighting	**0.178**$^{\uparrow\diamond}$	**0.190**$^{\uparrow\diamond}$	0.253	**0.146**$^{\uparrow\diamond}$	0.165$^{\uparrow}$	0.198	0.196	0.207	0.285
zero high-norm scores	0.152	0.168	0.241	0.121$^{\uparrow}$	0.148$^{\uparrow}$	0.186	0.199	0.208	0.285
Mean-map subtraction	0.138	0.165	0.208	0.077$^{\diamond}$	0.087$^{\downarrow\diamond}$	0.144$^{\diamond}$	0.201	0.252$^{\uparrow}$	0.233$^{\downarrow}$
subtract centre Gaussian	0.133	0.168	0.167$^{\downarrow}$	0.056$^{\downarrow}$	0.062$^{\downarrow}$	0.024$^{\downarrow}$	0.200	0.253$^{\uparrow}$	0.232$^{\downarrow}$
Probe relocation	0.041$^{\downarrow}$	0.084$^{\downarrow}$	0.076$^{\downarrow}$	0.027$^{\downarrow\diamond}$	0.050$^{\downarrow}$	0.051$^{\downarrow}$	0.084$^{\downarrow}$	0.143$^{\downarrow}$	0.141$^{\downarrow}$
matched blur	0.095$^{\downarrow}$	0.152	0.181$^{\downarrow}$	−0.008$^{\downarrow}$	0.068$^{\downarrow}$	0.097$^{\downarrow}$	0.189	0.226	0.287
All three corrections	0.059$^{\downarrow}$	0.123$^{\downarrow}$	0.076$^{\downarrow}$	0.079	0.137$^{\uparrow}$	0.085$^{\downarrow}$	0.088$^{\downarrow}$	0.177$^{\downarrow}$	0.090$^{\downarrow}$
+ test-time register	0.152	0.169	0.236	–	–	–	–	–	–
Top-k training (MiTA)	0.100$^{\downarrow}$	0.115$^{\downarrow}$	–	–	–	–	–	–	–
GAE (reference)	0.204	0.234	0.345	0.196	0.213	0.308	0.221	0.234	0.418

Execution test. The 14×14 grid is pooled into 16 overlapping regions. In every layer/head, each region's mean query selects its top-k keys; CLS retains its own set. Sets from an unmodified forward pass restrict every query to its region's set in all 12 layers simultaneously. Top-k sets beat equal-size random sets at every tested k (4–128) in all models, and agreement rises with k (Fig. B.1). The dotted attention-removed baseline alone uses uniform crop ratio 0.875; solid/dashed DeiT curves use 0.9.

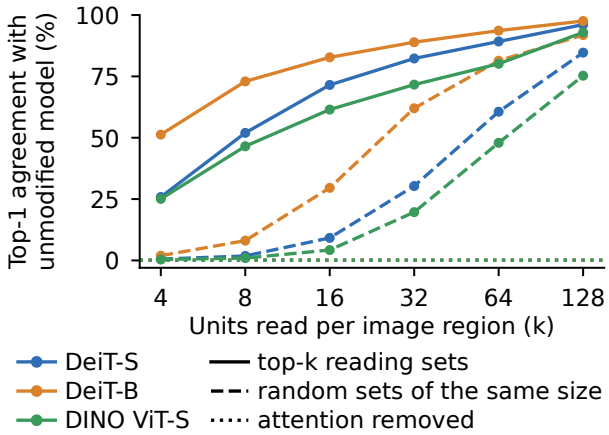


Figure B.1: Top-1 agreement on 5,000 ImageNet images with region-selected top-k units (solid), random units (dashed) or attention removed (dotted). Stored 95% intervals are at most ±1.5 percentage points. Dotted curves use crop ratio 0.875 for all models; other curves use each checkpoint's main preprocessing.

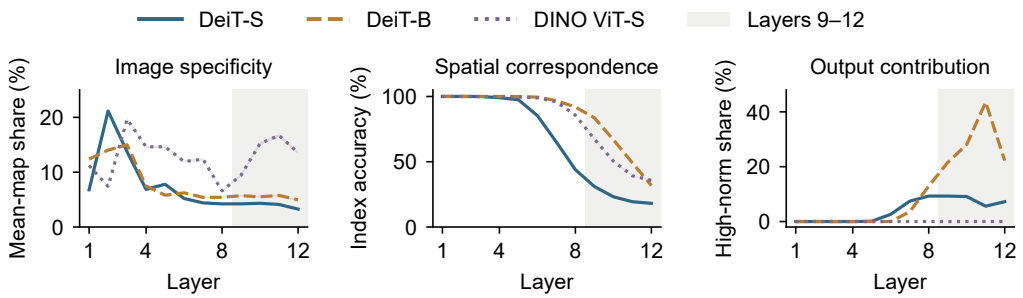


Figure B.2: Per-layer diagnostics: spatial variance explained by a held-out mean map, linear grid-index accuracy, and attention received by high-block-input-norm tokens. Curves have 95% intervals on ImageNet-S (left/right) or 2,000 held-out images (middle); shading marks layers 9–12. These measurements motivate tests, rather than establishing a causal link from diagnostic severity to map improvement.

Diagnostics per layer. Table B.2 lists the plotted diagnostics and uncorrected per-layer ClickMe alignment. In layers 9–12, 82–98% of mean-map variance is explained by distance from the grid centre. Earlier measurements on 11,919 ImageNet-S images used uniform crop ratio 0.875, rather than the main DeiT crop ratio 0.9: for each eligible image and head, we divided the mean raw value length $\|v_j^h\|$ of high-norm patches by the median for ordinary patches. Averaging these ratios within a layer and then equally over layers 9–12 gave 0.29 (DeiT-S) and 0.23 (DeiT-B); images without high-norm patches do not enter the ratio. This ratio was observed before the main registration. Block-input norms define the outlier group before LayerNorm; raw value lengths are measured after LayerNorm and the affine value projection. They measure different vectors. The ratios support downweighting these sources but do not measure their semantic information or final effect after vector cancellation and later computation.

Recovery at the last layer. In the supervised models, single-layer received attention is worse at layer 12 than at the best of layers 8–11 (ClickMe values in Table B.2). Applied to layer 12, value weighting recovers 62% and 33% of this drop on enrichment (DeiT-S, DeiT-B), 31% and 45% on ClickMe alignment, and 8% and 29% on deletion. The other two corrections make layer 12 worse.

Table B.2: Per-layer diagnostics (Fig. B.2) and uncorrected ClickMe alignment (last three columns). S, B, D: DeiT-S, DeiT-B, DINO ViT-S; probe chance: 0.5%. Values are rounded; DINO layers 9–12 have exactly zero high-norm attention on this dataset.

	Specificity: mean-map share (%)			Location: probe accuracy (%)			Contribution: high-norm share (%)			Map quality: ClickMe alignment		
Layer	S	B	D	S	B	D	S	B	D	S	B	D
1	6.8	12.4	11.2	100.0	100.0	99.9	0.0	0.0	0.0	−0.108	−0.111	−0.089
2	21.2	14.0	7.4	100.0	100.0	100.0	0.0	0.0	0.0	0.007	0.041	0.064
3	13.7	15.0	19.7	99.9	100.0	99.9	0.0	0.0	0.0	0.081	0.004	0.056
4	6.9	7.5	14.6	99.0	100.0	99.8	0.0	0.0	0.0	0.132	0.110	0.140
5	7.8	5.8	14.6	97.5	99.9	99.6	0.1	0.0	0.0	0.148	0.130	0.150
6	5.2	6.2	11.9	85.3	99.5	98.9	2.6	0.0	0.0	0.136	0.151	0.167
7	4.4	5.4	12.4	65.0	96.8	95.9	7.4	3.5	0.0	0.147	0.131	0.150
8	4.2	5.4	6.6	43.9	91.8	85.6	9.3	12.9	0.0	0.149	0.118	0.170
9	4.2	5.7	9.5	31.0	83.6	67.4	9.3	21.5	0.0	0.184	0.112	0.198
10	4.3	5.5	15.2	23.1	66.8	50.5	9.1	28.1	0.0	0.171	0.139	0.207
11	4.1	5.8	16.7	19.4	49.2	39.2	5.6	43.5	0.0	0.149	0.120	0.211
12	3.3	5.0	13.6	18.1	31.8	35.7	7.2	22.2	0.0	0.098	0.023	0.179

Table B.3: Paired differences, layers 9–12: mean, 95% image-bootstrap interval and verdict (*h*/*c*/*l*: higher/comparable/lower). Margins: 0.02 (enrichment/ClickMe), 0.05 (deletion); intervals unadjusted. Samples: App. A. Top: descriptive correction–R0 contrasts and the primary combination–R0 contrasts. Bottom: primary correction–control contrasts. Registered negative controls are also discussed in App. A.

Model	Same-ring enrichment	ClickMe alignment	Deletion score
Value weighting − received attention			
DeiT-S	0.038 [0.036, 0.039] *h*	0.033 [0.032, 0.034] *h*	0.017 [0.013, 0.021] *c*
DeiT-B	0.065 [0.063, 0.067] *h*	0.058 [0.056, 0.059] *h*	0.029 [0.024, 0.034] *c*
DINO ViT-S	−0.004 [−0.004, −0.003] *c*	−0.001 [−0.001, −0.001] *c*	0.000 [−0.003, 0.004] *c*
Mean-map subtraction − received attention			
DeiT-S	−0.002 [−0.003, −0.002] *c*	0.009 [0.008, 0.010] *c*	−0.028 [−0.034, −0.023] *c*
DeiT-B	−0.003 [−0.004, −0.003] *c*	−0.020 [−0.021, −0.019] *l*	−0.025 [−0.031, −0.019] *c*
DINO ViT-S	0.002 [0.001, 0.003] *c*	0.044 [0.040, 0.048] *h*	−0.052 [−0.059, −0.044] *l*
Probe relocation − received attention			
DeiT-S	−0.099 [−0.101, −0.097] *l*	−0.073 [−0.076, −0.069] *l*	−0.161 [−0.171, −0.150] *l*
DeiT-B	−0.054 [−0.055, −0.052] *l*	−0.057 [−0.059, −0.055] *l*	−0.118 [−0.127, −0.110] *l*
DINO ViT-S	−0.115 [−0.117, −0.113] *l*	−0.064 [−0.067, −0.062] *l*	−0.144 [−0.153, −0.134] *l*
All three corrections − received attention			
DeiT-S	−0.081 [−0.083, −0.079] *l*	−0.034 [−0.038, −0.030] *l*	−0.160 [−0.171, −0.149] *l*
DeiT-B	−0.001 [−0.003, 0.001] *c*	0.030 [0.027, 0.032] *h*	−0.084 [−0.093, −0.076] *l*
DINO ViT-S	−0.111 [−0.114, −0.109] *l*	−0.031 [−0.035, −0.027] *l*	−0.195 [−0.205, −0.184] *l*
Value weighting − zero high-norm scores (its control)			
DeiT-S	0.026 [0.025, 0.027] *h*	0.022 [0.021, 0.023] *h*	0.012 [0.008, 0.016] *c*
DeiT-B	0.025 [0.024, 0.026] *h*	0.017 [0.016, 0.018] *c*	0.013 [0.008, 0.018] *c*
DINO ViT-S	−0.004 [−0.004, −0.003] *c*	−0.001 [−0.001, −0.001] *c*	0.000 [−0.003, 0.004] *c*
Mean-map subtraction − centre-Gaussian subtraction (its control)			
DeiT-S	0.005 [0.004, 0.005] *c*	−0.003 [−0.005, −0.001] *c*	0.040 [0.034, 0.047] *c*
DeiT-B	0.022 [0.021, 0.023] *h*	0.025 [0.020, 0.030] *h*	0.120 [0.109, 0.130] *h*
DINO ViT-S	0.001 [0.001, 0.002] *c*	−0.001 [−0.001, 0.000] *c*	0.001 [−0.003, 0.005] *c*
Probe relocation − matched blur (its control)			
DeiT-S	−0.054 [−0.056, −0.051] *l*	−0.068 [−0.073, −0.064] *l*	−0.105 [−0.116, −0.095] *l*
DeiT-B	0.035 [0.034, 0.036] *h*	−0.018 [−0.020, −0.015] *c*	−0.047 [−0.055, −0.037] *c*
DINO ViT-S	−0.105 [−0.107, −0.102] *l*	−0.083 [−0.085, −0.080] *l*	−0.146 [−0.155, −0.137] *l*

Differences behind Table B.1. Table B.3 gives the paired differences with their intervals. The values in Table B.1 have 95% intervals of at most ± 0.005 on enrichment and ClickMe alignment and ± 0.012 on deletion. The model-side differences from DeiT-S received attention are $+0.012$ [0.011, 0.013] on enrichment, $+0.013$ [0.012, 0.013] on ClickMe and 0.000 [-0.005, 0.004] on deletion for the test-time register, and -0.040 [-0.042, -0.039] and -0.042 [-0.044, -0.040] on the first two for top-k training. The paired MiTA enrichment contrast uses 12,118 images valid for both models.

Tests of the contribution account. Value weighting reduces the relative weight of attention attached to small transformed-value vectors. One consequence was registered (P4a); additional checks were post hoc. *No practically meaningful gain in DINO (P4a).* In DINO ViT-S, with zero high-norm attention at layers 9–12 on ImageNet-S, every difference from received attention is at most 0.004 in size (Table B.3). *High-norm attention and alignment gain (post hoc).* Across images, gains have a weak positive association with the share of received attention on high-norm tokens: Spearman $+0.075$ [0.057, 0.092] and $+0.132$ [0.113, 0.150] on enrichment (DeiT-S, DeiT-B), $+0.045$ [0.032, 0.060] and $+0.158$ [0.143, 0.173] on ClickMe. *Assigned versus shuffled weights (post hoc).* Against the same weights shuffled among the patches of each head, value weighting is higher in all four supervised cells: $+0.038$ [0.037, 0.040] and $+0.073$ [0.071, 0.076] on enrichment, $+0.035$ [0.035, 0.036] and $+0.067$ [0.065, 0.068] on ClickMe. The shuffle control averages seeds 0–4 within each image before bootstrap resampling. Projected minus raw value weighting ranged from -0.0104 to -0.0004 across the six model–alignment comparisons. None reached the 0.02 margin; the study did not test equivalence. C3 aggregates per-head terms related to Kobayashi et al. [19] and shares a norm-weighted component with Zhu et al. [36]; their full analyses and aggregation differ.

B.1 Parameter-randomisation sanity check

The original study included a model-parameter randomisation check [2] on 1,000 fixed-seed ImageNet-S images. Parameters were cumulatively re-initialised from the classifier and final norm through blocks 12–1 to the patch/position embeddings and CLS token. The trained model's probes and mean maps remained fixed. Received attention and its corrections are class-agnostic and therefore do not depend on classifier parameters alone. Table B.4 reports the stored median per-image Spearman correlations between trained and fully randomised maps, with layers 9–12 summed. All five readings meet the registered median < 0.5 criterion; the model-independent centre Gaussian fails as designed. This is a necessary sensitivity check, not proof of prediction faithfulness. The existing summary is reproduced here without rerunning the experiment.

Table B.4: Stored model-randomisation results: median map correlation after full randomisation. The five readings pass the registered < 0.5 criterion in every model.

Reading	DeiT-S	DeiT-B	DINO ViT-S
Received attention	−0.033	0.055	0.071
Mean-map subtraction	0.187	0.366	0.100
Probe relocation	0.063	0.004	−0.002
Value weighting	−0.064	0.070	0.114
All three corrections	0.069	0.078	0.025
Centre Gaussian	1.000	1.000	1.000

C Tests of spatial relocation

Relocation lowered all nine deep-layer scores relative to received attention (Table B.1). Two planned comparisons narrow the problem. Early relocation remains comparable under the registered rule (P4b): at layers 1–4 it changes enrichment and ClickMe alignment by at most 0.004 in every model. In DeiT-S and DINO ViT-S, it also scored below blur matched for mean squared displacement (Table B.3). The rest of this section is post hoc: Table C.1 lists four explanations that we tested after seeing this result; none meets its specified criterion. Only 2–29% of the probe's error mass lay more than three patches from the true slot. Replacing the linear probe with the tested MLP did not restore alignment. The probe targets index decodability rather than input-content provenance; whether this mismatch explains the failed relocation remains untested (Section 3).

Table C.1: Four explanations for probe-relocation failure (post hoc). Model order: DeiT-S, DeiT-B, DINO ViT-S. Cells are the two centre-controlled metrics in the three models; criteria and probe settings are defined in this section.

Explanation	Test (layers 9–12 unless stated)	Result
The probe sends score to distant patches	Share of the probe's error mass more than three patches from the true patch	Small: 2–29% across models and layers. At layer 12, 45%, 66% and 81% of the probability lies in the 3×3 neighbourhood of the true patch
The probe has a fixed position preference	Peak of the probe's mean prediction relative to uniform; its correlation with the mean map	Close to uniform (peak 1.1–1.3 times the uniform value); at most weakly related to the mean map
The linear probe is too weak	Two-layer MLP probe in place of the linear one	Layer-12 accuracy rises by only 1.8, 3.8 and 6.7 points; relocation is still lower than received attention in all six cells
Global tokens are harder to place	Probe accuracy and entropy against a token's similarity to the image's mean token (layers 10–12)	No consistent relation; accuracy falls with globality only in DeiT-B

Probe capacity and explanatory criteria. The tested MLP has a 1,024-unit hidden layer, GELU and a 196-way output. It uses the linear probe's optimiser, epochs, minibatches and standardisation, trained on 1,800 of the original probe-training images with 200 for best-validation-epoch selection; the original 2,000 test images remain separate. After observing the original relocation result, we specified the following criteria before running these follow-up tests: error mass beyond three patches greater than 0.5 at every deep model/layer; fixed preference peak at least twice uniform and mean-map correlation at least 0.5 (with positive lower interval bound) in every deep model/layer; MLP layer-12 accuracy gain above 10 percentage points in some model, with relocation rescued if it is not lower in at least 4/6 alignment comparisons. None of these criteria was met. This MLP comparison leaves other probe designs and calibration errors open.

Globality analysis. Excluding high-norm tokens, globality is a token's cosine similarity to the mean of the other retained tokens in the image. Tokens are split into five equal-size rank groups. We relate the group rank to index accuracy and probe entropy, overall and within four equal-count radial groups, at layers 10–12. After observing the original relocation result, we set a follow-up criterion requiring negative accuracy and positive entropy associations (overall intervals and within-ring means) in at least two models. No consistent relationship meets this criterion.

D Example maps

We drew 24 ClickMe images at random with a fixed seed. Fig. D.1 shows the two with the smallest ClickMe index, and Fig. D.2 the next four. Received attention and value weighting look much alike, while probe relocation spreads the score over many more patches. Because each panel has its own colour scale, use Table B.1 to assess changes in patch ranking across images.

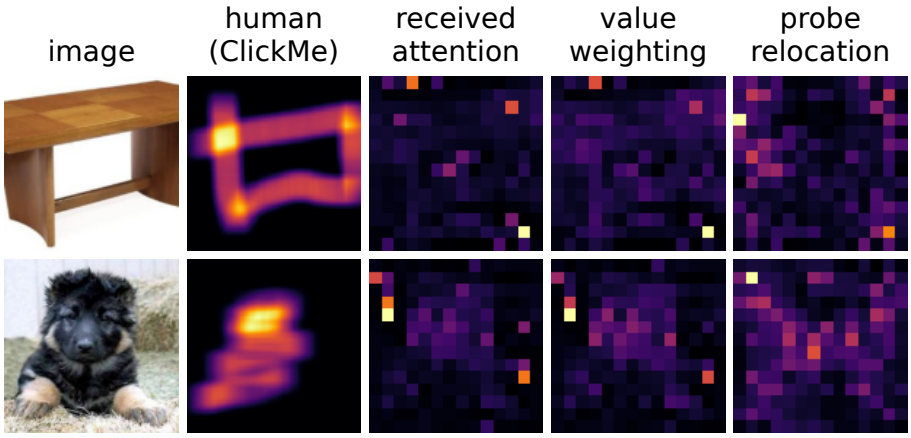


Figure D.1: The first two fixed-seed examples, DeiT-S, layers 9–12. Colours show scores min–max normalised within each map; amplitudes are not comparable across panels. These images were not selected for correction gain.

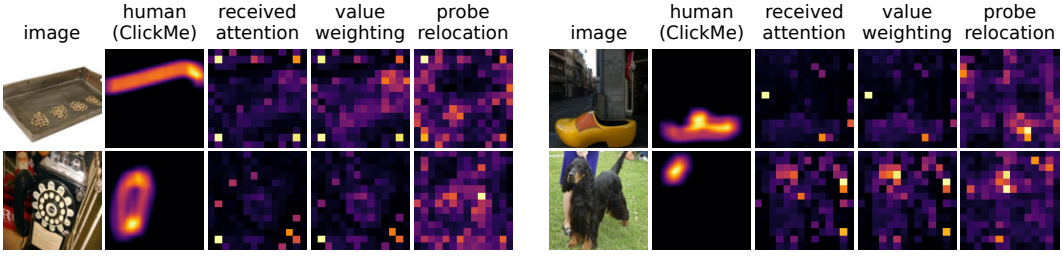


Figure D.2: DeiT-S on four further ClickMe images, layers 9–12; colours are normalised within each map and do not compare amplitudes across panels.

E Exploratory tests of spatial alignment and classifier sensitivity

Deletion endpoints. For each image and method, trapezoidal integration uses the original removal budgets, divided by 196. The probability endpoint integrates the probability of the original top-1 class under least-important-first removal minus most-important-first removal. The KL endpoint uses the reverse difference in divergence from the original prediction, so that larger scores again indicate greater separation. For both endpoints the paired curve difference is zero at budget zero. Results are paired over images. These measures have no registered practical margin.

Uncertainty. Differences are paired by image. Intervals use 1,000 image-bootstrap resamples with seed 0. ClickMe reuses each human map within a batch for its own and another image. Recomputing intervals by sampling the 272 original 64-image batches preserves those dependencies. For C3 relative to zeroing high-norm patches, the resulting ClickMe intervals are [0.021, 0.023] in DeiT-S and [0.016, 0.018] in DeiT-B. All estimates condition on the fixed within-batch pairing and the available checkpoints.

Table E.1: Exploratory endpoint differences (C3 minus comparator). Probability and KL columns give the mean paired area and 95% image-bootstrap interval over the fixed 2,000-image deletion subset. ClickMe uses the original 17,408 images with batch-bootstrap intervals. KL has units of nats times fraction removed.

Model	Comparator	Probability area	KL area	ClickMe
DeiT-S	R0	0.014 [0.012, 0.015]	0.058 [0.050, 0.066]	0.033 [0.032, 0.034]
DeiT-S	Zero high-norm	0.010 [0.009, 0.012]	0.041 [0.033, 0.049]	0.022 [0.021, 0.023]
DeiT-B	R0	0.022 [0.020, 0.024]	0.107 [0.096, 0.118]	0.058 [0.056, 0.059]
DeiT-B	Zero high-norm	0.010 [0.008, 0.012]	0.048 [0.038, 0.058]	0.017 [0.016, 0.018]
DINO	R0	0.000 [−0.001, 0.002]	−0.003 [−0.009, 0.003]	−0.001 [−0.001, −0.001]
DINO	Zero high-norm	0.000 [−0.001, 0.002]	−0.003 [−0.009, 0.003]	−0.001 [−0.001, −0.001]

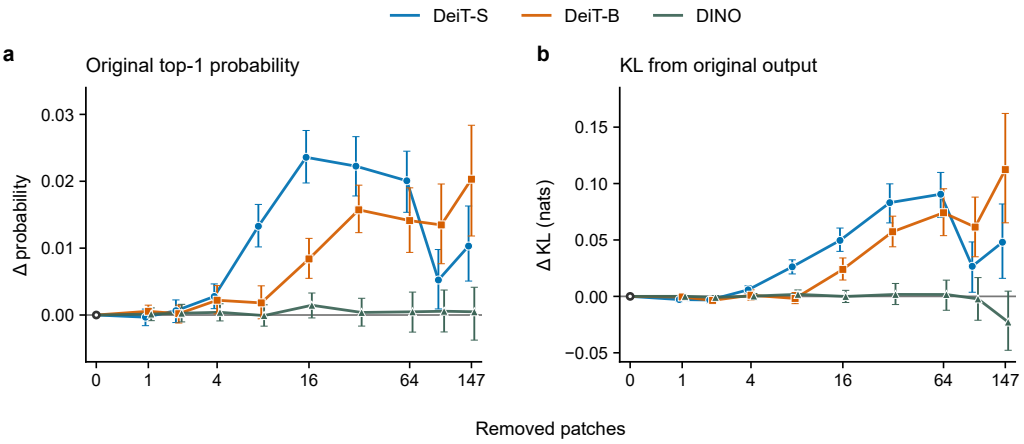


Figure E.1: C3 minus the zero-high-norm control at each original ImageNet-S deletion budget. The probability panel measures least-important-first minus most-important-first target probability. The KL panel reverses the subtraction to express divergence in the same direction. Horizontal positions use a log scale in one plus the number of removed patches, with model-specific offsets for visibility. Intervals are pointwise 95% paired image-bootstrap intervals over 2,000 images per model. The integrated advantage can coexist with a reversal at a small budget.

E.1 Object geometry and the enrichment scale

For an image with M valid patches and $k > 0$ foreground patches, let ring r contain m_r valid patches of which f_r are foreground. Selecting a patch in this ring contributes $(y - f_r/m_r)/k$ to enrichment, where y is its foreground indicator. Foreground contributions are nonnegative and background contributions are nonpositive. Selecting all k foreground patches therefore attains the maximum

$$E_{\max} = 1 - \frac{1}{k} \sum_{r:m_r>0} \frac{f_r^2}{m_r} \;\le\; 1 - \frac{k}{M}.$$

The inequality follows by Cauchy–Schwarz. It bounds the score itself. The possible change from R0 also depends on R0's starting score. A large raw improvement for small objects can thus reflect properties of both the map and the metric.

Area and centroid displacement. Object area is the share of valid patch slots labelled foreground. Centroid displacement is the distance from their centroid to the grid centre, in patches. For each checkpoint we computed the empirical quartiles on its finite-geometry images, assigning ties to the lower bin. The DeiT area cut points were 0.158, 0.342 and 0.571. DINO's different crop produced different masks. We paired scores within the same supplementary run and resampled images 2,000 times. In DeiT-S and DeiT-B the largest-minus-smallest area quartile differences in the C3 gain were -0.090 and -0.157 for enrichment. Comparing the most off-centre with the most central quartile instead gave $+0.057$ and $+0.095$. Area and centroid displacement covary. Among the smallest objects the latter differences were $+0.041$ [0.028, 0.053] and $+0.047$ [0.028, 0.066]. Some joint bins are small and these are observational contrasts.

Metric sensitivity. After seeing the enrichment gradients we examined patch AUROC under the same quartile cut points. AUROC is the probability that a foreground patch outranks a background patch, counting ties as one half. In the supervised models the gain in AUROC persists across area quartiles (Table E.2). AUROC does not condition on radius, so this comparison changes spatial adjustment as well as how foreground patches are selected. Overall C3-minus-R0 AUROC changes are $+0.028$, $+0.046$ and -0.005 for DeiT-S, DeiT-B and DINO.

Historical sample membership. Archived ImageNet-S files record zero enrichment for all-foreground grids, as the ring formula implies. Counts are 241 such grids for each DeiT and 265 for DINO. An additional 269/275 images, respectively, have zero object area and are excluded from the archived enrichment means. AUROC requires both classes, giving 11,909 DeiT and 11,879 DINO images. Restricting enrichment to that common eligibility changes C3-minus-R0 from 0.03757 to 0.03833 in DeiT-S, 0.06539 to 0.06671 in DeiT-B and -0.00351 to -0.00359 in DINO. The original verdicts use the archived cohort. In two DeiT-S images the original and supplementary R0 exports differ slightly. The stratified comparisons pair both methods from the supplementary export.

Table E.2: Mean C3 minus R0 by foreground-area quartile on ImageNet-S. Q1 contains the smallest objects. Both metrics use the same within-model cut points. AUROC excludes all-foreground crops.

Metric	Model	Q1	Q2	Q3	Q4
Enrichment	DeiT-S	0.092	0.041	0.013	0.002
Enrichment	DeiT-B	0.165	0.060	0.026	0.008
Enrichment	DINO	−0.003	−0.004	−0.005	−0.003
AUROC	DeiT-S	0.024	0.030	0.031	0.026
AUROC	DeiT-B	0.042	0.048	0.048	0.046
AUROC	DINO	0.000	−0.002	−0.006	−0.013

E.2 Independent pet-image controls

We hashed each Oxford test filename with the prefix `v13-spatial-2026:` and selected the first 384 in hash order. We resized the short side to 248 pixels and centre-cropped to 224, using bicubic interpolation for images and nearest-neighbour for trimaps. We ignored boundary pixels and excluded patches with no valid pixels. A patch was foreground if at least half its valid pixels were foreground. One crop contained only foreground. Adjacent images in hash order exchanged length ranks, preserving each recipient's lengths within each layer and head. Within-ring shuffles used the same permutation across heads for a given image and layer. The reverse control (`norm_only`) keeps projected lengths at their own patches. Within each image, layer and head it replaces received attention at every patch with the same mean over the 196 patch slots. This preserves each head's total patch attention but removes its spatial variation.

We evaluated all seven readouts on the same 383 eligible crops. Table 1 reports paired differences in the primary endpoint, same-ring enrichment; the table below gives corresponding differences in AUROC. Both use 95% percentile intervals from 10,000 paired image-bootstrap draws (seed 2026), without a practical margin. A follow-up sensitivity check resampled the 192 donor pairs, weighting by the number of eligible images in each draw. C3 minus donor-rank gives 0.0223 [0.0187, 0.0261] in this check. The captured C3 map and logits were checked against the original implementation on the first batch. Each query–patch pair satisfied the triangle bound within numerical tolerance over the run.

Table E.3: All seven spatial readouts on the same 383 pet crops. Each AUROC difference subtracts the AUROC of value weighting at the original positions. Mean attention preserves each head's total patch attention; net output sums head vectors before taking each query–source magnitude, then pools queries.

Reading	AUROC	ΔAUROC [95% CI]	Enrichment
Received attention	0.788	−0.028 [−0.031, −0.026]	0.141
Value weighting (own lengths)	0.817	—	0.160
Donor-image positions	0.787	−0.029 [−0.032, −0.026]	0.137
Within-ring permutation	0.793	−0.023 [−0.025, −0.021]	0.138
Own lengths, mean attention	0.757	−0.059 [−0.069, −0.049]	0.120
Unprojected value norm	0.822	0.006 [0.005, 0.006]	0.163
Norm after summing heads	0.809	−0.008 [−0.009, −0.007]	0.156

E.3 Last-block CLS value intervention

A second fixed filename hash selected 128 of the 384 images independently of their scores. At block 12 we cached its input and the unmodified logits. For each patch j we held the original CLS attention weights fixed and removed $t_j = \sum_h a^h_{\mathrm{CLS},j} z^h_j$ from the CLS attention output. The shared projection bias stayed in place. The remaining residual, token-wise MLP, final normalisation and classifier were evaluated for each modified CLS. If m is the margin between the original highest and second-highest logits, the response was $|m_{-j} - m|$ for those same classes. This intervention concerns the contextualised token at that slot. Its input evidence can span multiple image regions.

For each image, we correlated the 196 responses with each unmodified layer-12 readout by Spearman correlation. The means over images were 0.696 (all-query R0), 0.705 (all-query C3), 0.730 (CLS attention), 0.747 (CLS sum of head magnitudes) and 0.756 (CLS net magnitude). The paired CLS-only sum versus all-query C3 difference was 0.041 [0.026, 0.057] with 10,000 image-bootstrap resamples. Explicit interventions in full forward passes on four images and three patch indices each agreed with vectorised logits to within 2.4×10^{-6}. Cached inference took about 3 seconds on an RTX 4060; individual patch responses are in `experiments/cls/`.

Matched mask comparison. After inspecting the classifier intervention we evaluated the saved layer-12 rankings on the same 128 images and masks. All 128 had eligible foreground and background patches. Table 2 reports the three outcomes with equal image weights. Keeping the CLS query fixed, moving from the sum of per-head magnitudes to net magnitude changed margin Spearman by $+0.0092$ [0.0069, 0.0116], enrichment by $+0.0037$ [0.0007, 0.0069] and AUROC by $+0.0002$ [-0.0031, 0.0033]. The intervention measures sensitivity to a contextualised source in this layer, not to input pixels.

E.4 Receiver comparison in DeiT-B

In DeiT-S, restricting the readout to CLS improved intervention ranking but reduced foreground AUROC. We tested whether both directions persisted in a second checkpoint. Before scoring DeiT-B, we specified both endpoints in `experiments/receiver_review/`. We retained the same 128 pet images, preprocessing, block 12, five readouts, 196 interventions per image and bootstrap settings. All 128 crops contained both mask classes. This follow-up was exploratory relative to the main registration.

The checkpoint is `deit_base_patch16_224.fb_in1k`, revision `b78cc5532a69df6bcad9c3a8d76653fd20b31ac6`. Independent full-model forwards checked four images at patches 0, 97 and 195. Maximum absolute logit and fixed-class-margin errors were 2.86×10^{-6} and 1.91×10^{-6}, below the specified 0.002 tolerance. The run, including checkpoint loading, took 73 seconds on an RTX 4060 Laptop GPU (PyTorch 2.6.0+cu124). Scores, selected filenames, source hashes, validation rows and metadata are saved separately under `experiments/receiver_review/deit_b/`; the original DeiT-S archive remains unchanged.

Table E.4: All five layer-12 rankings on the same 128 images. Intervention ρ measures agreement with CLS-path sensitivity; AUROC and enrichment compare the unmodified score maps with foreground masks.

Model	Readout	Intervention ρ	Mask AUROC	Enrichment
DeiT-S	All queries, attention	0.696	0.754	0.117
DeiT-S	Value weighting, all queries	0.705	0.769	0.127
DeiT-S	Value weighting, CLS only	0.747	0.678	0.095
DeiT-S	CLS, net vector	0.756	0.678	0.098
DeiT-S	CLS, attention	0.730	0.666	0.081
DeiT-B	All queries, attention	0.589	0.649	0.051
DeiT-B	Value weighting, all queries	0.652	0.685	0.084
DeiT-B	Value weighting, CLS only	0.734	0.710	0.088
DeiT-B	CLS, net vector	0.751	0.719	0.096
DeiT-B	CLS, attention	0.675	0.676	0.062

Table E.5: Paired effects of retaining only the CLS query in the sum of head magnitudes. Intervals are 95% percentile intervals from 10,000 image-bootstrap draws, seed 2026; no registered practical margin is applied.

Model	Endpoint	CLS − all queries [95% CI]
DeiT-S	Intervention ρ	0.041 [0.026, 0.057]
DeiT-S	Mask AUROC	−0.091 [−0.110, −0.074]
DeiT-S	Enrichment	−0.033 [−0.043, −0.023]
DeiT-B	Intervention ρ	0.082 [0.068, 0.096]
DeiT-B	Mask AUROC	0.024 [0.013, 0.036]
DeiT-B	Enrichment	0.005 [−0.005, 0.013]

The intervention ranking improves in both models, as expected from the target's CLS construction. The mask comparison differs: CLS-only reduces AUROC by 0.091 in DeiT-S but raises it by 0.024 in DeiT-B. DeiT-B enrichment changes by 0.005 with an interval containing zero. These results do not support a universal localisation–classifier-sensitivity trade-off. They show that the receiver choice must be checked against the intended endpoint for the particular checkpoint. Retaining cancellation between heads is another distinction: in DeiT-B, net magnitude versus summed head magnitudes changes intervention correlation by 0.017 [0.014, 0.020], AUROC by 0.009 [0.007, 0.011] and enrichment by 0.008 [0.005, 0.011].

Scope across datasets. The receiver comparison covers only the pet images. The stored ImageNet-S map scores cannot reconstruct the CLS interventions, and the raw images and masks were unavailable for this extension.

F Key parts of the code

Given the attention a_{ij}^h and values v_j^h of a layer, the readings are:

```
def received_and_contribution(p, v, w_o, prefix=1):
    # p: [B,H,T,T]; v: [B,H,T,d]; w_o: [D,H,d]
    # T includes CLS; values include the value-projection bias.
    gram = torch.einsum('dhe,dhf->hef', w_o, w_o)
    length = torch.einsum('bhne,hef,bhnf->bhn', v, gram, v)
    length = length.clamp_min(0).sqrt()
    received = p.sum(2)  # sum over queries
    s = received.sum(1)[:, prefix:]
    c = (received * length).sum(1)[:, prefix:]
    return s, c

def relocate(s, probs):  # source j to predicted position p
    return torch.einsum('bj,bjp->bp', s, probs)
```

For the independent pet experiment, the following readouts vary the placement of lengths or remove spatial variation in received attention. Here a has axes (image, head, query, patch), and z contains each head's projected value vector. The implementation processes queries in chunks.

```
# Own lengths versus the same lengths in a donor image's spatial order
length = z.norm(dim=-1)                          # [B, H, 196]
order = donor_length.argsort(dim=-1)
borrowed = torch.empty_like(length).scatter(-1, order, length.sort(-1).values)
received = a.sum(dim=2)
own_map = (received * length).sum(dim=1)
donor_map = (received * borrowed).sum(dim=1)
mean_attention_map = (received.mean(dim=-1, keepdim=True) * length).sum(dim=1)

# Magnitude after summing the head vectors for each query and patch
net_map = sum(torch.einsum('bhij,bhjd->bijd', chunk, z).norm(dim=-1).sum(1)
              for chunk in a.split(32, dim=2))
```

For the final-block intervention we hold the attention matrix fixed. This excerpt from the full-forward check removes one patch's term from the CLS query. The output projection retains its bias and the rest of the model runs normally.

```
# p: [batch, head, query, source]; v: [batch, head, source, head_dim]
head_out = p @ v
head_out[:, :, 0, :] -= p[:, :, 0, source_patch + 1, None] * v[:, :, source_patch + 1, :]
full = head_out.transpose(1, 2).reshape(batch, tokens, -1)
return attn.proj_drop(attn.proj(attn.norm(full)))
```

AI Use Statement. I set the direction of this project, chose a fast-weight perspective for studying ViT attention maps, and organized the paper around three assumptions about how these maps are interpreted. I used AI tools to help search the literature, design follow-up experiments, write code, run experiments, analyze results, and draft and revise the report.